



Set Interface from Collection:

What is Set?

- Set in java collection framework is a collection of unique elements, where duplicate elements are not allowed.
- The set interface contains only methods inherited from collection and adds the restriction that duplicate elements are prohibited.
- Set interface is the child interface of collection.
- Set interface is found in **java.util** package.

Object Creation of Set Interface:

```
Set<Obj> set = new HashSet<Obj>( );
```

Where: - Set → Interface
 <Obj> → Generic (inside angular bracket)
 set → Reference variable / object
 new → Keyword
 HashSet → Implemented Class
 () → Parentheses

Note – Obj is the type of the object to be stored in Set.



Difference between List and Set:

There are following difference between list and set which is given below –

List Interface	Set Interface
List allows duplicates values	Set can't have duplicate values
Lists can be indexed and positionally accessed.	Sets cannot be indexed and cannot be accessed based on their positions.
Multiple null elements can be stored.	Only one null element can be stored at a time.
List implementations are ArrayList, LinkedList and Vector etc.	Set implementations are HashSet, LinkedHashSet and TreeSet etc.
The insertion order is maintained by the list.	It doesn't maintain the insertion order of elements.
List is often used when we need to frequently access the elements using their indices.	It is used when we need to store distinct elements.



What is an Enhanced For Loop in Java?

- This loop provides way for traversing the array or collections and hence it is mainly used for traversing array or collections.
- It was introduced in java 1.5 and it is also a control flow statements that iterates a part of the program multiple times.
- This loop also makes the code more readable.

Syntax of an Enhanced For Loop:

```
for (datatype item : array){  
    }  
}
```

Here → **array** – An array or a collection.

item – Each item of array/collection is assigned to this variable.

datatype – The data type of the array/collection